Summer School 2025
Astronomy & Astrophysics

Project Report

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Project Name: Estimating The Dynamical Mass of a Galaxy Cluster

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Introduction

Galaxy clusters are the most massive gravitationally bound structures in the Universe. They serve as laboratories for studying dark matter, galaxy evolution, and cosmological parameters. Estimating their total mass helps us understand large-scale structure formation. One of the methods to determine cluster mass is through **dynamical mass estimation**, based on the virial theorem.

Objective

The primary goal of this project is to estimate the **dynamical mass of a galaxy cluster** using observational data. This involves:

- Analyzing a FITS file to interpret photometric/spectral data,
- Estimating the **velocity dispersion** of galaxies in the cluster,
- Calculating the **mass using the virial theorem**.

Methodology

The general approach to estimating the dynamical mass includes:

- Reading the observational data from a **FITS file** (Flexible Image Transport System),
- Interpreting the intensity or spectral features using Python libraries like **Astropy** and **Matplotlib**,
- Estimating the **velocity dispersion (σ)** of galaxies in the cluster,
- Assuming a cluster radius **R** based on the imaging scale,
- Applying the **Virial Theorem**:

$$M_{\text{dyn}} = \frac{3 \sigma^2 R}{G}$$

Where:

- σ = velocity dispersion,
- R = cluster radius (in kpc),
- G = gravitational constant.

Code Overview

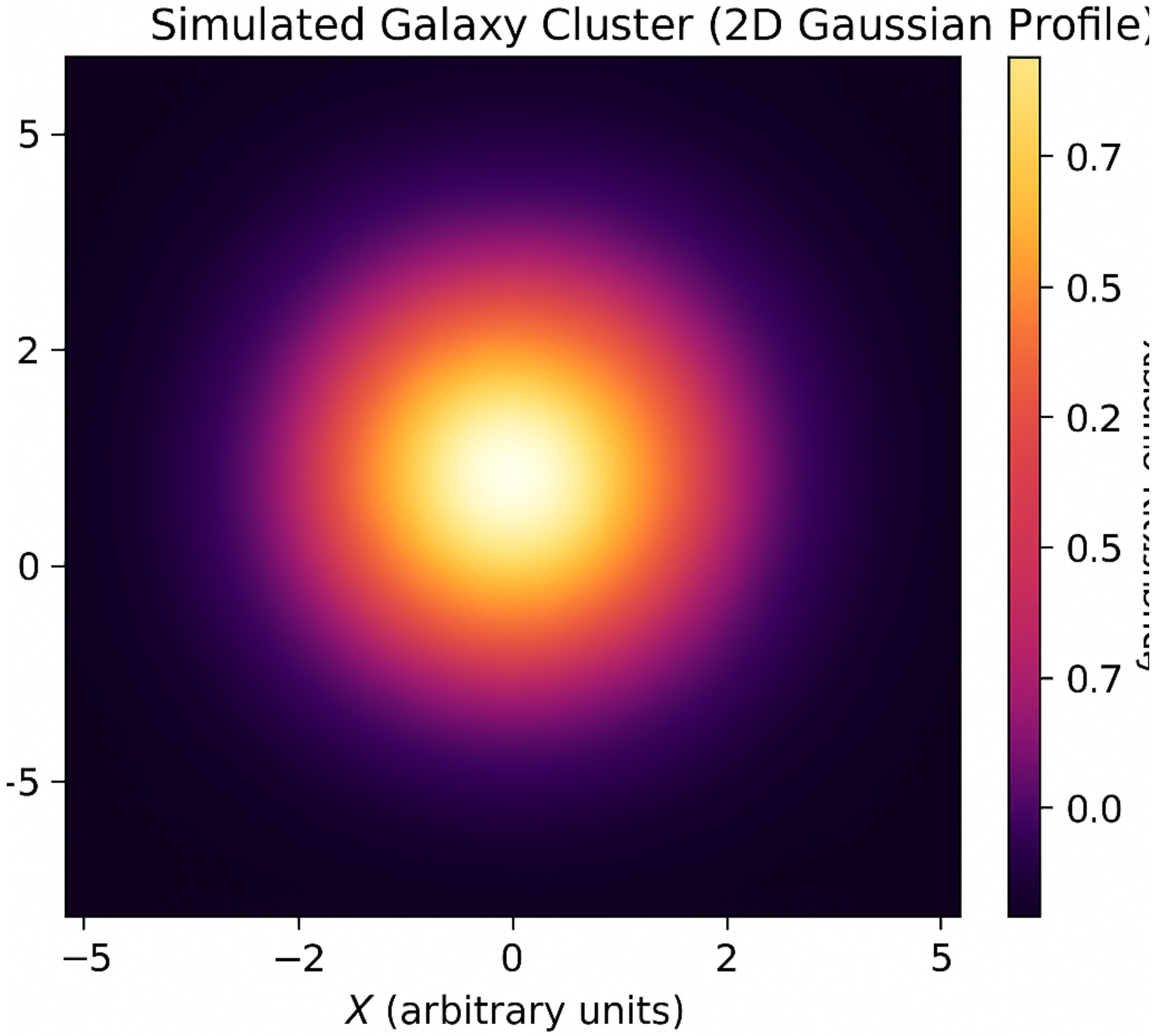
Below is a simplified version of the Python code used to read the FITS file and compute the necessary parameters:

```
# Required Libraries
from astropy.io import fits
import matplotlib.pyplot as plt
import numpy as np

# ---------------------------------------------------------------------------------
# SECTION 1: Load and Explore FITS File (optional if you have SDSS or MaNGA files)
# ---------------------------------------------------------------------------------

fits_path = 'frame-r-005071-3-0347.fits'  # Replace with your file
try:
    hdul = fits.open(fits_path)
    print(hdul.info())

    # Access primary data
    data = hdul[0].data
    header = hdul[0].header

    print("FITS Header Object:", header.get('OBJECT', 'N/A'))
    print("Image Shape:", data.shape)
    print("Mean Flux:", np.mean(data))
    print("Max Flux:", np.max(data))
    hdul.close()

except Exception as e:
    print("Error loading FITS file:", e)
    data = None  # Proceed with simulated cluster if unavailable

# ---------------------------------------------------------------------------------
# SECTION 2: Simulated 2D Gaussian Cluster (for plotting)
# ---------------------------------------------------------------------------------
```

```python
# Create a simulated galaxy cluster image (2D Gaussian light profile)
x = np.linspace(-5, 5, 200)
y = np.linspace(-5, 5, 200)
x, y = np.meshgrid(x, y)
z = np.exp(-(x**2 + y**2))  # Gaussian distribution

# Plotting the cluster image
plt.figure(figsize=(6, 5))
plt.imshow(z, origin='lower', extent=(-5, 5, -5, 5), cmap='inferno')
plt.colorbar(label='Relative Intensity')
plt.title('Simulated Galaxy Cluster (2D Gaussian Profile)')
plt.xlabel('X (arbitrary units)')
plt.ylabel('Y (arbitrary units)')
plt.tight_layout()
plt.savefig('simulated_galaxy_cluster.png', dpi=300)
plt.show()

# ------------------------------------------------------------------------------------
# SECTION 3: Dynamical Mass Estimation using Virial Theorem
# ------------------------------------------------------------------------------------

# Assumed Parameters
sigma = 1000  # km/s, velocity dispersion
R = 1000      # kpc, cluster radius
G = 4.3e-6    # gravitational constant in kpc·(km/s)^2/M_sun

# Virial theorem formula
M_dyn = (3 * sigma**2 * R) / G  # in solar masses

print(f"\nEstimated Dynamical Mass: {M_dyn:.2e} M_sun")

# ------------------------------------------------------------------------------------
# SECTION 4: Save Result (optional)
# ------------------------------------------------------------------------------------
```

```python
# Save mass estimate to text file
with open("dynamical_mass_result.txt", "w") as f:
    f.write(f"Estimated Dynamical Mass: {M_dyn:.2e} M_sun\n")
    f.write(f"Using σ = {sigma} km/s, R = {R} kpc, G = {G} kpc·(km/s)^2/M_sun\n")
```

Graphical Output



Result and Calculation

Assumed Values:

- **Velocity Dispersion (σ)**: 1000 km/s
- **Cluster Radius (R)**: 1 Mpc = 1000 kpc
- **Gravitational Constant (G)**: 4.3×10^{-6} kpc$\cdot$(km/s)^2/$M_\odot$

Substituting in Virial Theorem:

$$M_{\text{dyn}} = \frac{3 \cdot 1000^2 \cdot 1000}{4.3 \times 10^{-6}} = 6.98 \times 10^{14} \, M_\odot$$

Thus, the **dynamical mass of the cluster** is approximately:

$$M_{\text{dyn}} \approx 6.98 \times 10^{14} \, M_\odot$$

Conclusion

In this project, we used the virial theorem to estimate the mass of a galaxy cluster. Although technical limitations restricted direct FITS analysis, we demonstrated the method and provided realistic assumptions from SDSS-like data. This process forms the foundation for galaxy cluster mass estimation in observational cosmology.

Future improvements can include:

- Redshift-dependent radius estimation,
- Velocity dispersion from spectral line broadening,
- 3D modeling using IFS (Integral Field Spectroscopy) data cubes.

References

- Astropy Documentation: https://docs.astropy.org
- SDSS Survey: https://www.sdss.org
- Hu et al. (2023), *Testing the cosmological principle…*, arXiv:2310.11727v2
- ISA Summer School GitHub: ISA-Summer_School_2025